VASA PREVIA

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OUTLINE

- Introduction
- Epidemiology
- Risk factors
- Pathogenesis
- Classification
- Diagnosis
- Management
- Summary

INTRODUCTION

- Vasa previa is defined as the presence of **fetal vessels over or adjacent to (within 2 cm) the cervical os.**

EPIDEMIOLOGY

- The overall incidence of vasa previa is **1 in 2500** deliveries.

RISK FACTORS

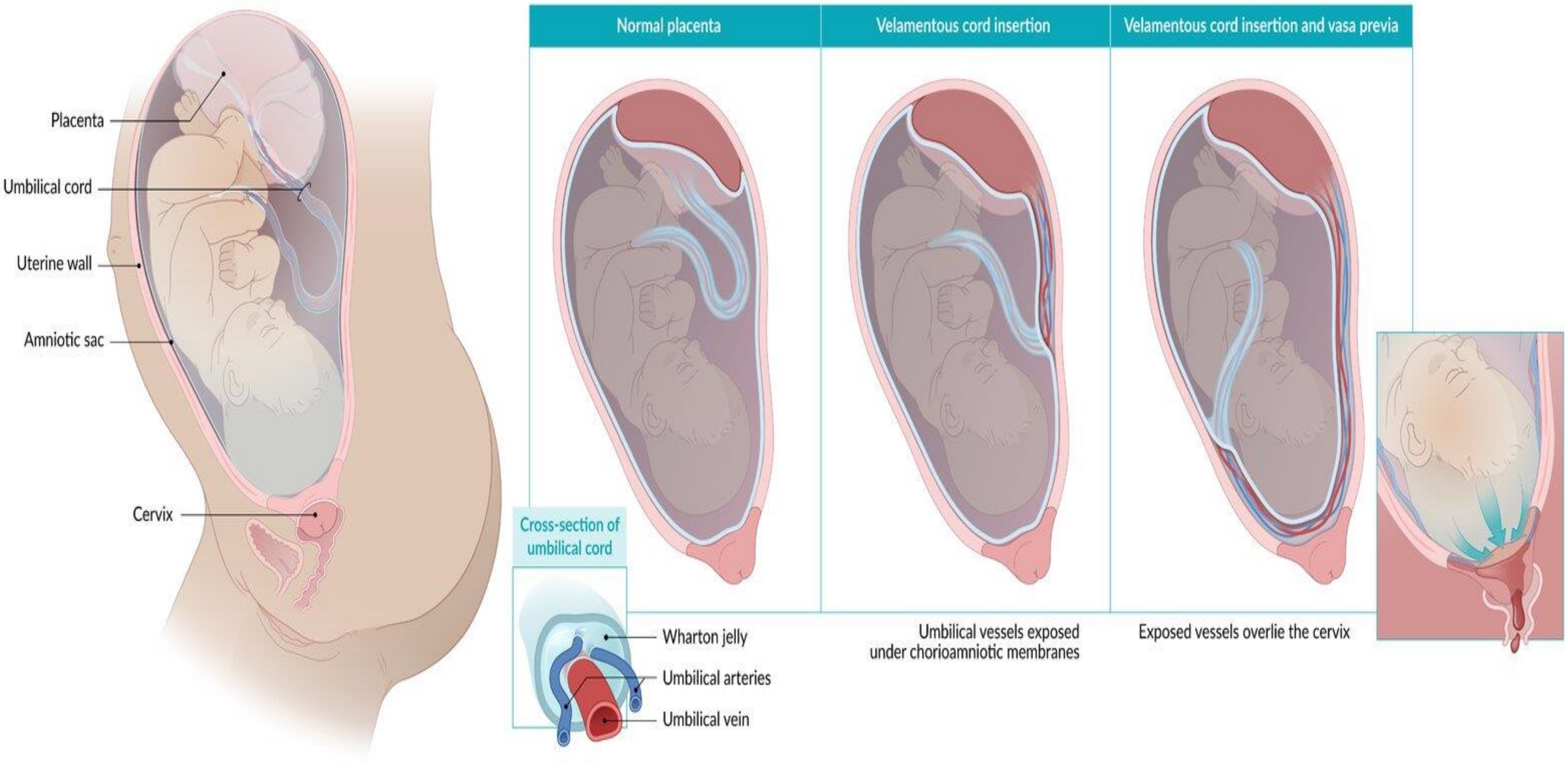
- Reported risk factors for vasa previa include:
  - Velamentous cord insertion,
  - Bilobed or succenturiate-lobed placentas,
  - Assisted reproductive technology,
  - Multiple gestations, and
  - Hx of 2^{nd}-TM placenta previa or low-lying placenta.

PATHOGENESIS

- Typically, **the vessels travel within the amniotic membrane** and lack protection from Wharton jelly.
- Because of this, the vessels are **vulnerable to rupture** & compression.
- When the vessels rupture, the fetus is at high risk for **exsanguination**.

CLASSIFICATION

- **Three-types** of vasa previa have been described:
  1. **Type 1**, in which the vessels arise from a **velamentous/marginal** cord insertion;
  2. **Type 2**, in which the membranous vessels connect a **bilobed or succenturiate-lobed** placenta;
  3. **Type 3**, in which one or more large vessels **run through the membranes along the margin of the placenta**, such as with a <u>resolving placenta previa</u>.



Placenta
Umbilical cord
Uterine wall
Amniotic sac
Cervix
Normal placenta
Velamentous cord insertion
Velamentous cord insertion and vasa previa
Cross-section of umbilical cord
Wharton jelly
Umbilical arteries
Umbilical vein
Umbilical vessels exposed under chorioamniotic membranes
Exposed vessels overlie the cervix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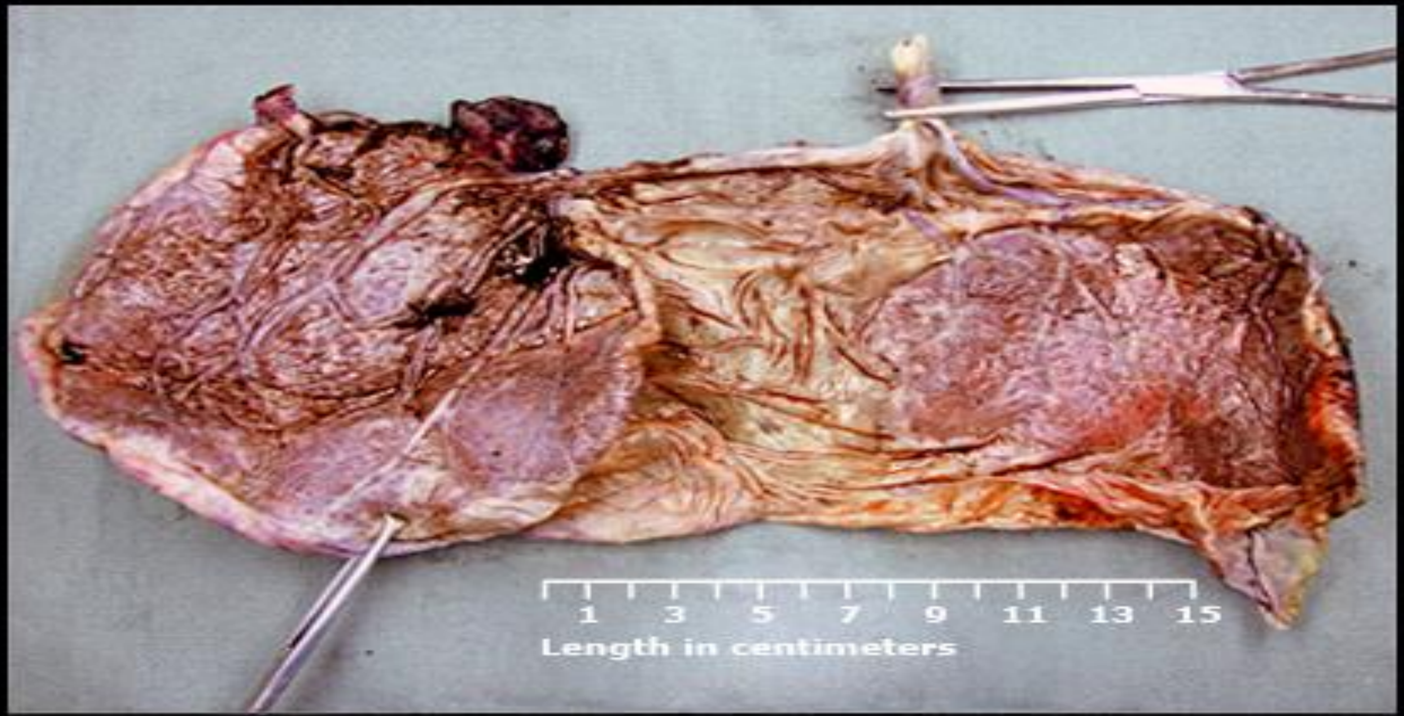
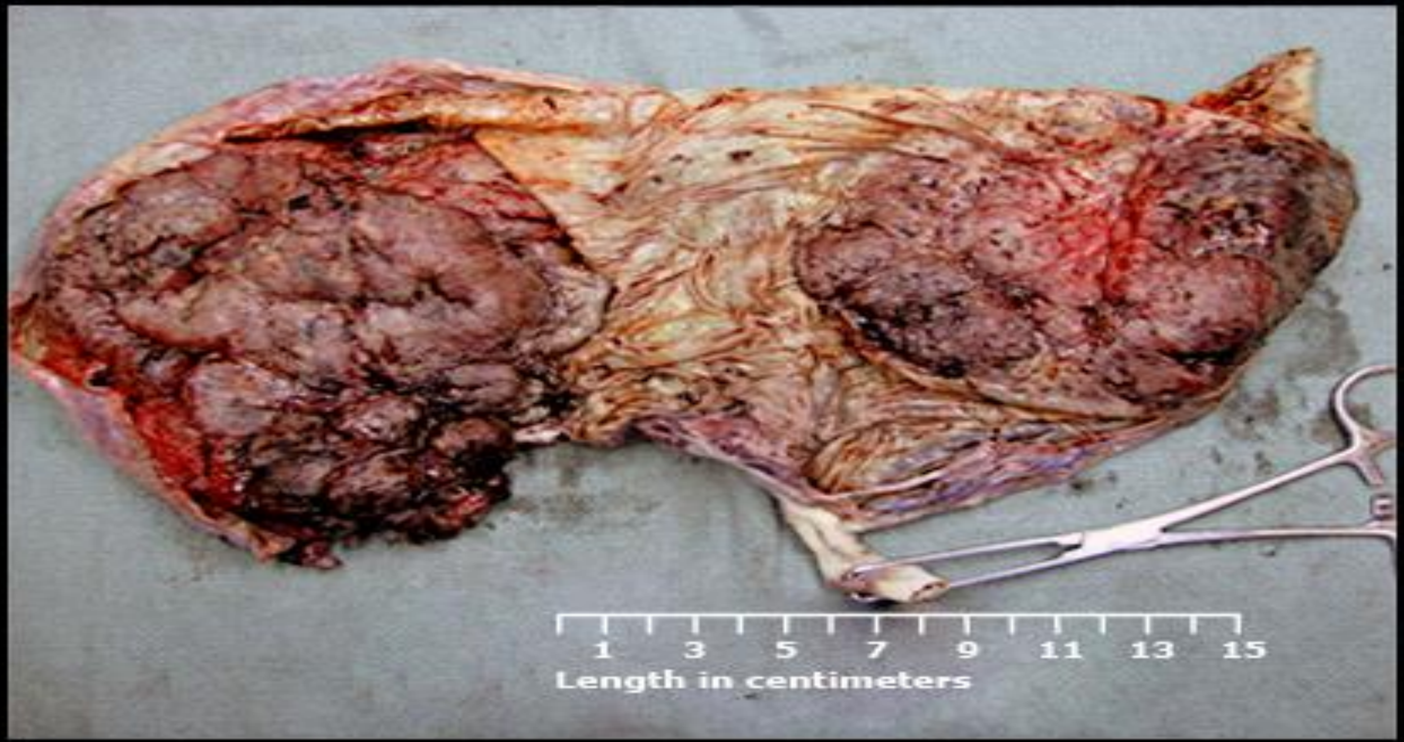

1 3 5 7 9 11 13 15
Length in centimeters


1 3 5 7 9 11 13 15
Length in centimeters

CLINICAL MANIFESTATIONS

- **In the past**, most cases of vasa previa presented after membrane rupture with acute vaginal bleeding from a lacerated fetal vessel and associated fetal bradycardia.

- **Today,** most cases of vasa previa are diagnosed antenatally by ultrasound.

- **In rare cases**, pulsating fetal vessels may be palpable in the membranes that overlie the cervical os.

DIAGNOSIS

- Vasa previa is often diagnosed antenatally by ultrasound with **color & pulsed Doppler** mapping.
- **Transabdominal and transvaginal** approaches are most typically used.
- The diagnosis is confirmed by documenting **umbilical vessels over the cervical os** using color and pulsed Doppler imaging

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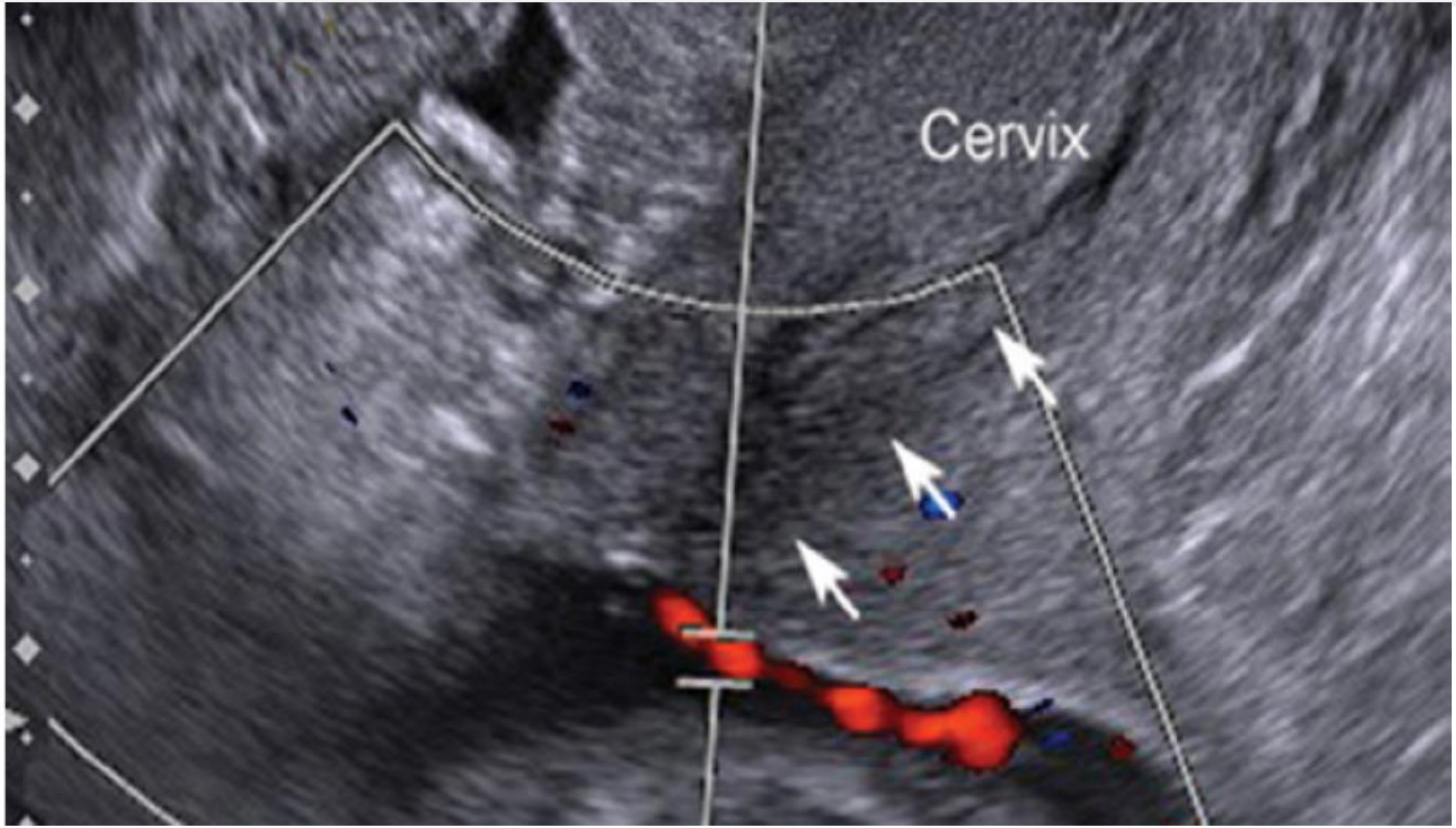


- An umbilical vessel (red linear structure) is seen
- Overlying the internal os and cervical canal (arrows).

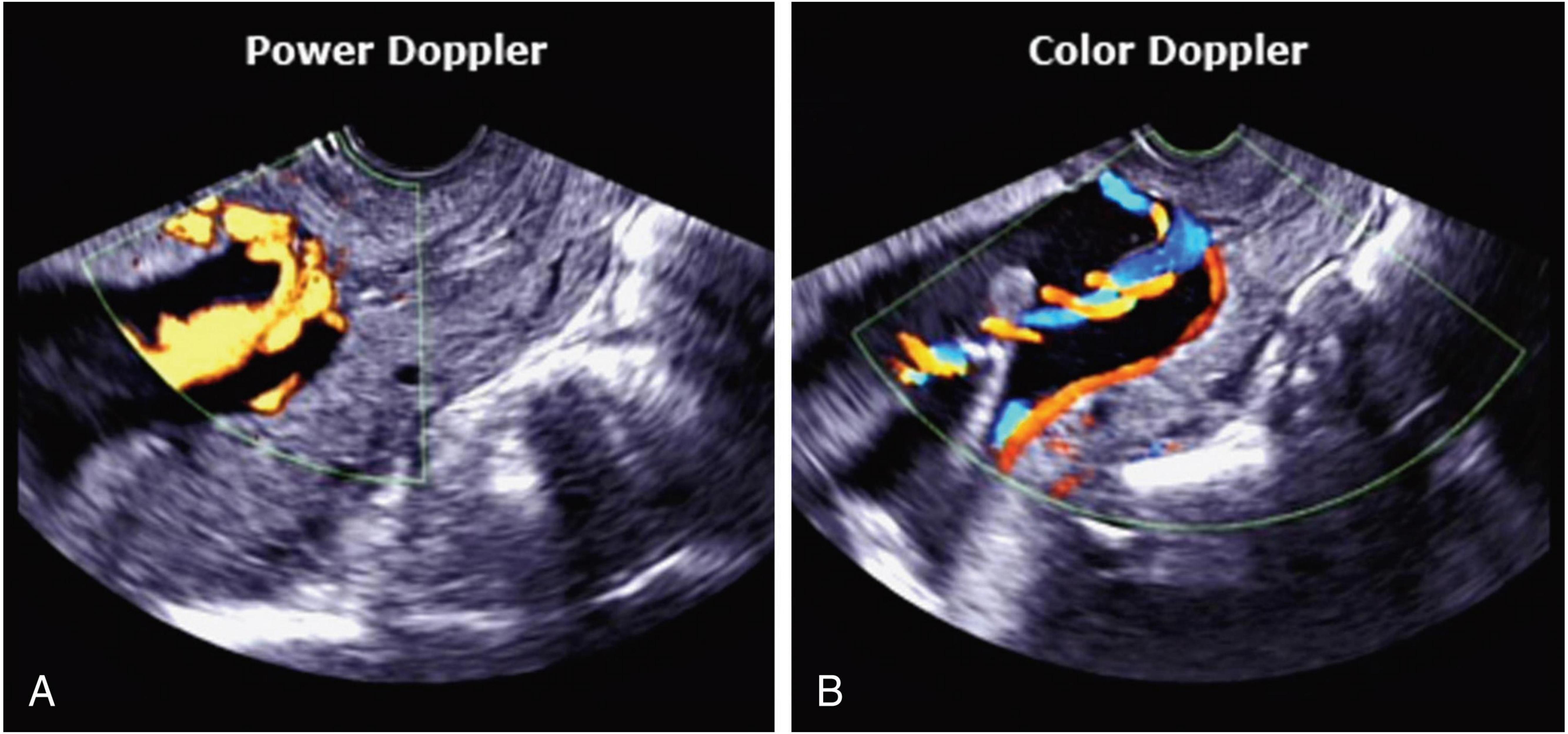


Fig. 18.10 **Transvaginal Ultrasound Images Showing Vasa Previa and Velamentous Cord.** The placenta is posterior, with an anterior succenturiate lobe. (From Lockwood CJ, Russo-Steiglitz K. *Velamentous umbilical cord insertion and vasa previa*, www.uptodate.com, May 27, 2018.)

MANAGEMENT

- When diagnosed antenatally, vasa previa is managed **similarly to placenta previa.**
- In pregnancies with vasa previa, **the goal** is
  - To deliver the fetus before ROM or onset of labor
  - To minimize the sequelae accruing from iatrogenic preterm birth.
- This decision is based on patient-specific factors and clinical judgment.

CONT'D

- When diagnosed antenatally, vasa previa is managed **similarly to placenta previa.**
- **Antenatal testing** is begun at 32 weeks to assess for fetal cord compression.
- **ACS** are administered prior to 34 weeks of gestation, and
- **$MgSO_4$** neuroprotection is provided if delivery under 32 weeks is anticipated.
- **Cesarean delivery** @**34—37** weeks of gestation pending clinical course.
  - **For asymptomatic patients** with no evidence of cervical shortening or preterm labor., *Outpatient management* is reasonable
  - **For symptomatic patients** with vaginal bleeding, uterine contractions, cervical shortening, or nonreassuring fetal testing, *inpatient management* is recommended.
- **If an intrapartum dx of vasa previa is made**, expeditious delivery is needed.
  - **Immediate neonatal blood transfusion** is often required in these circumstances.

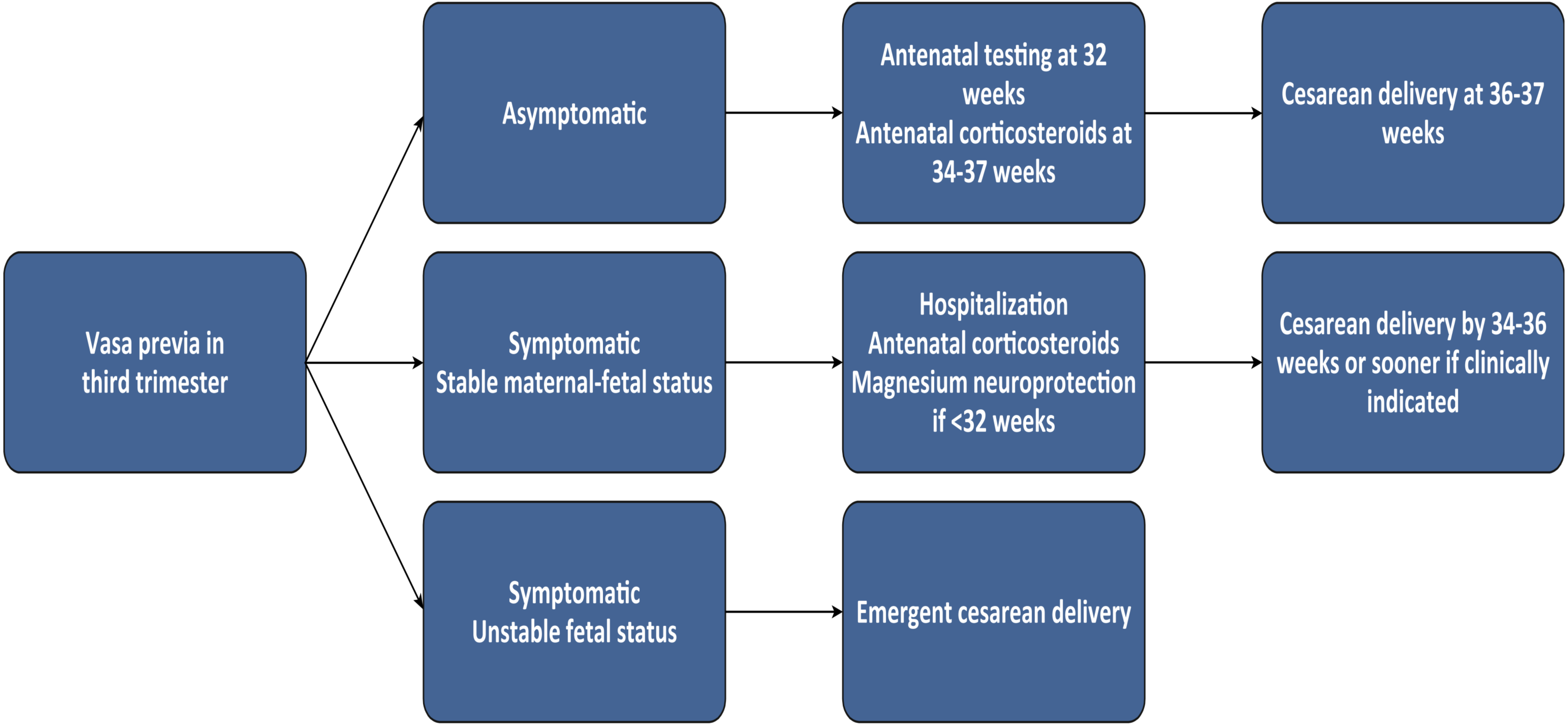
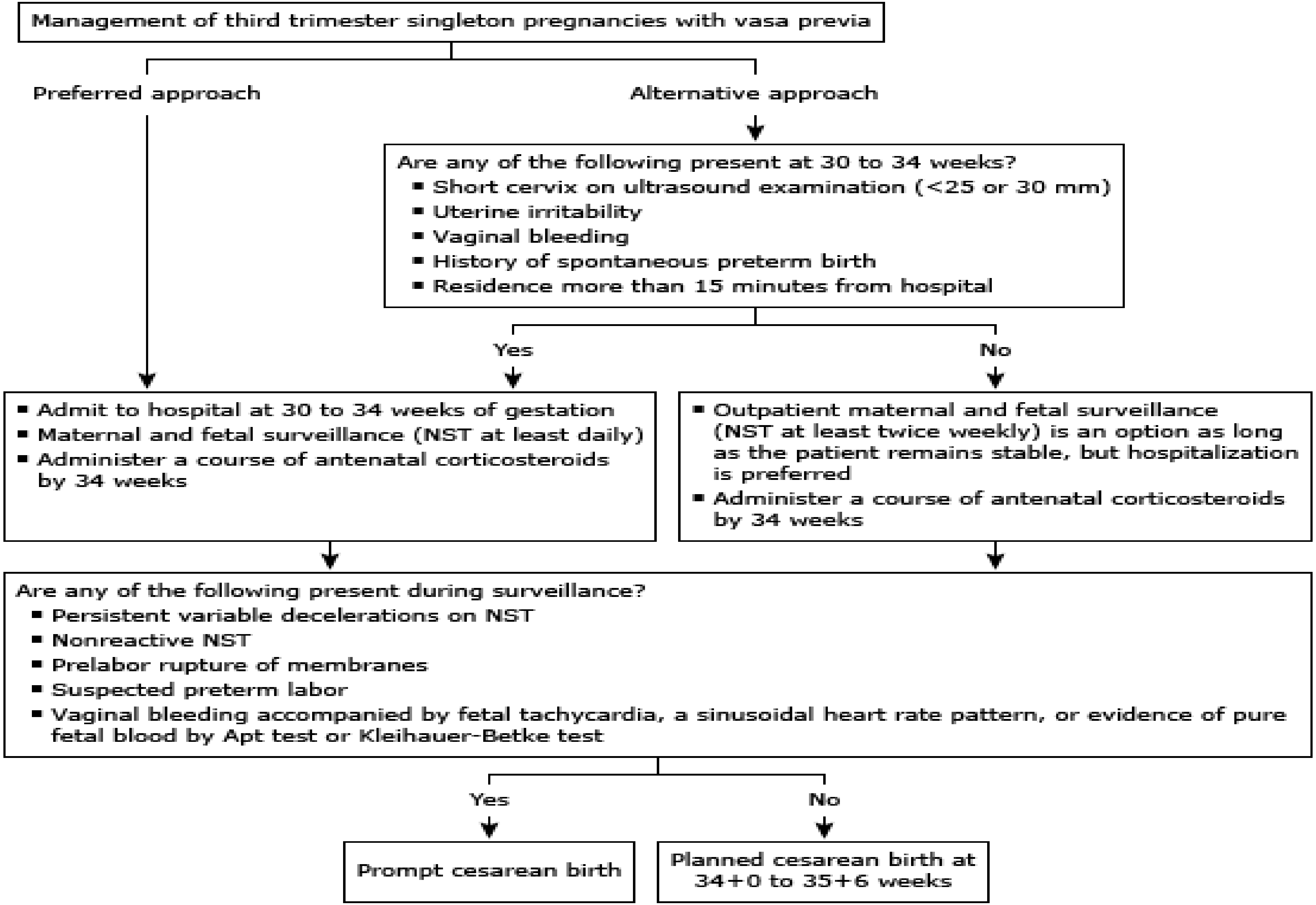


Fig. 20.11 Algorithm for Management of Vasa Previa.


Management of third trimester singleton pregnancies with vasa previa
Preferred approach
Alternative approach
Are any of the following present at 30 to 34 weeks?
▪ Short cervix on ultrasound examination (<25 or 30 mm)
▪ Uterine irritability
▪ Vaginal bleeding
▪ History of spontaneous preterm birth
▪ Residence more than 15 minutes from hospital
Yes
No
▪ Admit to hospital at 30 to 34 weeks of gestation
▪ Maternal and fetal surveillance (NST at least daily)
▪ Administer a course of antenatal corticosteroids by 34 weeks
▪ Outpatient maternal and fetal surveillance (NST at least twice weekly) is an option as long as the patient remains stable, but hospitalization is preferred
▪ Administer a course of antenatal corticosteroids by 34 weeks
Are any of the following present during surveillance?
▪ Persistent variable decelerations on NST
▪ Nonreactive NST
▪ Prelabor rupture of membranes
▪ Suspected preterm labor
▪ Vaginal bleeding accompanied by fetal tachycardia, a sinusoidal heart rate pattern, or evidence of pure fetal blood by Apt test or Kleihauer-Betke test
Yes
No
Prompt cesarean birth
Planned cesarean birth at 34+0 to 35+6 weeks

SUMMARY

Definition:

- Vessels travel within the membranes and overlie the cervical os
- They can be torn with cervical dilation or membrane
- Rupture, and laceration can lead to rapid fetal exsanguination.

Classification:

- **Type 1**: Vessels are part of a velamentous cord insertion
- **Type 2**: Vessels span b/n portions of a bilobate or succenturiate placenta

Diagnosis:

- Antepartum dx greatly improves the perinatal survival rate, which ranges from 97-100 %
- Mid trimester sonographic examination. In suspicious cases, **TVS.**
- Subsequent imaging is reasonable because up to 39% of cases ultimately resolve

Treatment(Hospitalization)

- Antenatal corticosteroid administration
- For surveillance & expedited delivery for labor, bleeding, or ROM.
- **Cesarean delivery @34-36 weeks of GA (ACOG)**

Vasa previa

Definition	• Fetal vessels overlying the cervix
Risk factors	• Placenta previa • Multiple gestations • In vitro fertilization • Succenturiate placental lobe
Clinical presentation	• Painless vaginal bleeding with ROM or contractions • FHR abnormalities (eg, bradycardia, sinusoidal pattern) • Fetal exsanguination & demise
Management	• Emergency cesarean delivery

REFERENCES

- Williams Obstetrics 26^{th} edition
- Gabbe's Obstetrics 9^{th} edition

THANK YOU